

COLLEGE ADMISSION → ENROLLMENT → GRADUATION → TOR

Expanded Development Guide

End-to-end blueprint for building, securing, deploying, and operating a college student lifecycle, academic records, enrollment, grading, graduation, Transcript of Records, and credential system.

Scope: Academic lifecycle + registrar + secure backend + workflows + integrations + infrastructure + operations + disaster recovery + production readiness.

Core principle: Preserve authoritative academic history, separate official records from working/planning data, and enforce authorization at the backend/database layer.

Configure institutional rules, curricula, grading policies, document formats, retention requirements, privacy requirements, and applicable Philippine higher-education requirements from approved institutional sources.

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1. Purpose, Scope & Development Principles

This guide extends the original College Admission-to-TOR blueprint into a broader technical development guide. It covers the academic lifecycle from application through enrollment, instruction, grades, graduation and TOR, plus the backend, security, infrastructure, integration and operational capabilities required for a production system.

Authoritative-source: The registrar/authorized academic record is authoritative for official enrollment, grades, credits, completion and credentials.

Historical-record: Do not silently overwrite meaningful academic history. Preserve attempts, status changes, curriculum versions, corrections and approved transactions.

Workflow: Important changes should use request → validation → approval → posting → audit.

Security: Enforce least privilege, authentication, role-based access and database-level authorization. Frontend visibility rules are not sufficient security.

Configurability: Do not hard-code institutional grading scales, prerequisites, academic-standing thresholds, honors rules, deadlines, document requirements or graduation rules.

Operations: Design for backup, restoration, monitoring, migration and business continuity before production launch.

2. Complete Student Lifecycle

PRE-ADMISSION → APPLICATION → DOCUMENT VERIFICATION → EXAM/INTERVIEW → ADMISSION DECISION → ACCEPTANCE → STUDENT NUMBER/ACCOUNT → ENROLLMENT ELIGIBILITY → ADVISING → COURSE SELECTION → PREREQUISITE/COREQUISITE CHECK → SECTION/SCHEDULE CHECK → APPROVAL → ASSESSMENT/PAYMENT → OFFICIAL REGISTRATION → SEMESTER DELIVERY → ATTENDANCE/ASSESSMENTS → GRADES → ACADEMIC STANDING → SEMESTER AUDIT → NEXT TERM → INTERNSHIP/OJT → RESEARCH/THESIS/CAPSTONE → GRADUATION APPLICATION → DEGREE AUDIT → CLEARANCE → GRADUATION APPROVAL → DEGREE CONFERMENT → TOR/DIPLOMA/CERTIFICATIONS → LICENSURE/EMPLOYMENT/FURTHER STUDY → ALUMNI.

Every transition should have a defined status, permitted actions, responsible role, validation rules, timestamps and audit trail.

3. System Modules

Module	Primary responsibility
Student Master Record	Identity, institutional identifiers, program, cohort, status, contacts
Admissions	Applications, requirements, exams, interviews, decisions
Document Management	Files, verification, versions, retention and release
Transfer Credit	Prior coursework, equivalencies, credit decisions and residency
Programs & Curriculum	Programs, curriculum versions, courses, prerequisites, substitutions
Academic Policy Engine	Configurable grading, standing, repeats, honors and graduation rules
Academic Advising	Plans, recommendations, approvals and degree audit
Academic Calendar	Years, terms, deadlines and grading windows
Enrollment	Registration, course selection, eligibility and official registration
Sections & Scheduling	Sections, instructors, rooms, capacity and timetable conflicts
Attendance & Assessment	Attendance and assessment records
Grades	Submission, approval, posting and corrections
Academic Standing	GPA, deficiencies and policy-based status
Finance & Scholarships	Charges, payments, holds, aid and scholarships
Student Services	ID, library, support and controlled clearances
Internship/OJT	Placement, agreements, hours, evaluations and completion
Research/Thesis/Capstone	Proposal, adviser, panel, defense, revisions and completion
Graduation	Application, degree audit, clearance and conferment
Registrar	Official academic history, corrections and certifications
TOR & Credentials	Generation, authentication, release and verification
Alumni	Post-graduation records and services

Module	Primary responsibility
Security & Audit	Authentication, authorization, logs and monitoring
Notifications	Email/SMS/in-app messages and delivery status
Integrations/API	Payment, SSO, email, SMS, external systems and webhooks
Operations	Backups, monitoring, deployment, recovery and support

4. Student Master Record

Keep stable identity information separate from transactional records.

Field group	Examples
Identity	Student ID, legal name, preferred name, date/place of birth
Contact	Email, mobile, address, emergency contact
Academic identity	Program, major, campus, college, cohort, curriculum version
Admission	Admission type, admission term, application reference
Status	Current status plus effective-dated history
Accounts	Portal identity, university email reference
Audit	Created/updated timestamps, actor, record version

5. Admissions & Document Management

Admission: Draft → Submitted → Document Check → Exam/Interview → Evaluation → Accepted/Waitlisted/Denied → Confirmation → Enrollment Eligibility.

Track application category, programs, requirements, evaluation results, conditions, deadlines, communications and scholarship/aid applications.

Documents have independent status and history: Required → Submitted → Verified or Rejected/Needs Correction → Replaced/Archived.

Separate document metadata from binary files. Track owner, type, version, verification actor, timestamp, expiration/retention and access history.

6. Transfer Credit & Advanced Standing

Never replace prior-course evidence with an institutional course. Preserve source course data and the approved equivalency decision.

Track prior institution, source course, units, grade, evidence, equivalent course, credit decision, approver, date and residency implications.

7. Programs, Curriculum & Academic Policy Engine

Model programs independently from curriculum versions. Associate students with the applicable curriculum version.

Entity	Important data
Program	Code, title, degree, department, duration, campus
Curriculum version	Version, effective term/year, status, cohort
Course	Code, title, units/credits, type, description
Curriculum course	Required/elective, sequence, year/term placement
Prerequisite/corequisite	Course relationships and conditions
Substitution	Original requirement, substitute, reason, approver, effective date
Transition rule	Old/new curriculum, cohorts, conditions

Policy engine: Configure grading scales, prerequisite exceptions, repeats, residency, honors, academic standing, graduation requirements, deadlines and document requirements.

8. Academic Advising & Degree Audit

The degree audit should answer what is completed, currently enrolled, remaining, permitted next, and deficient.

Include general education, major/core, electives, PE/NSTP or institutional requirements, internship, research/thesis/capstone, free electives, total units, residency and other graduation conditions.

For reproducibility, save curriculum version, ruleset/version and calculation date for official audits.

9. Academic Calendar & Term Management

Make the academic calendar a first-class module.

Object	Examples
Academic year	2026–2027
Term	Semester 1, Semester 2, Summer/Special Term
Enrollment window	Opening/closing timestamps
Add/drop window	Deadline
Withdrawal window	Deadline and approvals
Assessment periods	Prelim/midterm/final or institution-defined
Grade submission	Instructor and approval deadlines
Correction period	Permitted correction window
Graduation period	Application, audit and conferment dates

Use calendar data to enforce deadlines consistently.

10. Enrollment, Sections & Scheduling

Enrollment: term setup → eligibility → advising → selection → prerequisite/corequisite validation → capacity → schedule conflict → approval → assessment/payment → official registration → registration document.

Section data should include course, code, instructor(s), room, schedule, modality, capacity, enrollment, waitlist and status.

Scheduling should validate student, instructor and room conflicts. Larger institutions may use a timetable optimization service.

11. Add/Drop, Withdrawal & Exceptional Paths

Event	Typical workflow
Add course	Request → validation → approval if required → registration
Drop course	Request → deadline check → update registration
Change section	Request → capacity/conflict → approval/posting
Withdrawal	Request → policy/deadline → approval → transcript/financial effect
Leave	Request → review → approval → effective term → return/readmission
Readmission	Application → checks → approval → enrollment eligibility
Shifting	Request → evaluation → curriculum/credit mapping → approval
Transfer out	Request → clearance → official records/document release

12. Attendance, Assessment & Grade Management

Keep working teaching data separate from the official academic record.

Grade lifecycle: Working → Submitted → Reviewed/Approved → Posted/Official → Corrected only through controlled correction.

Corrections preserve original/replacement values, reason, evidence, requester, approver and timestamps.

Incomplete, failed, repeated, withdrawn, deferred and missing-grade statuses must be configurable.

13. Academic Standing & Progress Monitoring

Support the institution's approved GPA/standing model. Do not hard-code grading scale, rounding, repeat or honors policy.

Preserve every course attempt and maintain standing history with term, ruleset, calculated values, decision and authorized override.

14. Finance, Holds & Scholarships

Keep financial transactions separate from academic grades while allowing controlled eligibility checks.

Track charges, payments, references, balances, holds, releases, scholarships, conditions and renewals.

Production finance should add payment integration, receipts, reconciliation, refunds/voids and payment audit trails.

15. Student Services

Services may include ID, library, required clearances, organizations, support, career services and portal/email accounts. Integrate through controlled identifiers.

16. Internship / OJT / Practicum

Track eligibility, host, agreement, supervisor, dates, hours, attendance, evaluations, completion evidence and academic result.

17. Research / Thesis / Capstone

Topic → adviser → proposal → approval → research/development → defense → revisions → final submission → completion.

Track adviser, panel, dates, outcomes, revisions, files and repository references.

18. Graduation & Degree Conferment

Graduation is an auditable degree-completion process, not simply years attended.

Check curriculum, units, prerequisites, residency, internship, research/capstone, clearances and other approved conditions.

Record application, audit version, deficiencies, approvals, conferment date, degree, major and honors/distinctions where applicable.

19. Registrar Master Record

The registrar layer is authoritative for official academic history and credential generation.

Record	Examples
Identity	Institutional identifiers and verified identity
Enrollment	Official term registration/status
Academic history	Courses, units, grades, attempts, withdrawals
Program history	Program changes and curriculum versions
Standing	Official academic-standing history
Completion	Degree audit and graduation status
Credentials	TOR, certifications, diploma references
Corrections	Approved changes, reason, evidence, actor, timestamp

20. TOR, Credentials & Verification

TOR: Request → identity verification → eligibility/hold checks → registrar processing → academic record assembly → validation → official generation → authentication/security → release → release log.

Generate TORs only from authoritative records. Use controlled document numbers/reference numbers and, where approved, QR/online verification.

Credential corrections preserve the prior issuance and link replacement credentials to the correction event.

21. Alumni & Post-Graduation

Retain only information the institution is authorized and needs to keep. Do not overwrite academic history.

Optional services include licensure/board tracking, employment/graduate-school requests and credential verification.

22. Secure Backend Architecture

Recommended baseline: PostgreSQL-backed system with a controlled server-side application/API layer, dedicated authentication, private document storage, database-level authorization and auditable privileged operations.

Reference: user portals → application/API → authentication + RBAC → workflow services → PostgreSQL + RLS → private storage; notifications, payments, SSO and other integrations connect through controlled APIs/webhooks.

Critical: Frontend role checks are not sufficient. Sensitive operations must be authorized at the API/server and database layers.

A Supabase + PostgreSQL implementation can use Auth, PostgreSQL RLS, private Storage and server-only privileged operations; Django + PostgreSQL is another valid implementation when more custom control is required.

23. Authentication, RBAC & Database Authorization

Support secure login, recovery, session management, optional SSO and MFA for privileged accounts.

Permissions should distinguish view, create, submit, approve, post, correct, release and administer.

Use PostgreSQL Row Level Security or equivalent to restrict records by role, user, organization/campus/department and legitimate scope.

Never expose service credentials or privileged database keys to browsers.

Add rate limiting, secure sessions/tokens, login monitoring, recovery controls and security alerts.

24. API & Application Architecture

Use a clear boundary: UI → API/controller → authorization → validation → business logic → transaction → database.

Validate on the server. Do not trust client-supplied IDs, roles, grades, prices, approval states or document permissions.

Use transactions for official registration, grade posting and credential issuance. Version external APIs and maintain integration/error logs.

25. Workflow & Approval Engine

Create a reusable workflow engine for request → review → approval → posting.

Object	Purpose
Workflow definition	Reusable process and transitions
Workflow instance	Specific request/process
Step	Current stage
Assignment	Responsible role/user
Decision	Approve, reject, return, cancel, escalate
Comment/evidence	Reason/supporting information
Deadline	Expected completion
Delegation	Approved reassignment
Audit	Complete action history

Use it for admissions, transfer credits, substitutions, leave, readmission, shifting, grade corrections, graduation, TOR corrections and credential release.

26. Document & File Storage

Separate document metadata from file storage. Use private storage by default.

Track file version, checksum/hash where appropriate, owner/student, type, uploader, verification status, timestamps, retention date and release/download history.

Consider malware scanning, file-type restrictions, size limits and safe filename handling.

27. Notifications & Communications

Build email/SMS/in-app notification services with templates, queues, delivery status, retry handling and logs.

Examples: missing documents, application status, enrollment deadlines, grade posting, clearance deficiencies, graduation status and TOR readiness.

28. Payments & External Integrations

Possible integrations: payment gateways/banks, SSO, email, SMS, library, LMS and credential/verification services.

Use stable identifier mappings. Webhook handlers must be idempotent. Maintain integration health and reconciliation reports.

29. Database Architecture

Conceptual entities include: students, student_contacts, student_status_history; applications, requirements, decisions, application_events; document_metadata/document_versions; programs, curriculum_versions, courses, curriculum_courses, prerequisites, corequisites, substitutions, policy_rules; academic_years, terms, calendar_events; instructors, faculty_assignments, rooms, room_resources, sections, schedules, waitlists; enrollments, enrollment_items, enrollment_changes; attendance, assessments, grades, grade_changes, academic_standing; transfer_credits, program_changes; financial_accounts, charges, payments, refunds, holds, scholarships; student_services; internships/evaluations; research_projects/panels/documents; graduation_applications, degree_audits, clearances, degrees_conferred; tor_requests, credentials, credential_verifications; workflow_definitions/instances/actions; notifications/deliveries; users, roles, permissions, role_assignments; audit/security logs; API clients, integration/webhook events; reports; retention_policies/legal_holds.

Key chain: Student → Enrollment → Enrollment Item → Section → Course; Student → Program/Curriculum; Course → Prerequisites; Student → Degree Audit → Graduation → Degree Conferment → Credential/TOR.

30. Data Quality, Integrity & Concurrency

Use uniqueness, required-field, foreign-key and valid-range constraints.

Prevent duplicate identities through matching/review workflows.

Use transactions and constraints to prevent double enrollment, over-capacity, duplicate credentials, duplicate payments and conflicting updates.

Create reconciliation reports for enrollment, grades, finance, graduation and credentials.

Corrections should use controlled workflows, not direct database editing.

31. Audit Logging & Record History

Use security/system audit and academic-record audit concepts.

Academic audit should capture actor, role, action, entity, old/new values or references, reason, approval reference and timestamp; IP/user-agent may be retained where justified.

Protect logs from ordinary modification. High-value events include grade changes, curriculum changes, transfer-credit decisions, program shifts, graduation, degree conferment, TOR generation, credential release and permission changes.

32. Privacy, Retention & Data Governance

Define collection purpose, access scope, retention and disposal for personal, academic, financial, document and log data.

Support controlled correction/export/disclosure processes and legal/administrative holds where applicable.

Minimize sensitive data in reports, APIs, logs and notifications.

Validate final implementation against the institution's applicable Philippine privacy and higher-education requirements.

33. Backup, Disaster Recovery & Business Continuity

Define RPO and RTO for the system and critical subsystems.

Back up database and important files. Test restoration, not merely backup creation.

Consider point-in-time database recovery and independent document recovery.

Document outage procedures, communication, downtime transactions, reconciliation and return-to-service approval.

Run periodic recovery exercises.

34. Monitoring, Operations & DevOps

Monitor uptime, errors, database health, API latency, storage, queue failures, authentication anomalies and integrations.

Use controlled centralized logs; never place secrets or unnecessary student data in logs.

Separate development, staging/test and production. Keep secrets out of source code. Use reviewed deployment and rollback processes.

Monitor backups, certificates, domains and external dependencies.

35. Non-Functional Requirements

Requirement	Guidance
Security	Least privilege, MFA, encryption, RLS, auditability
Performance	Defined response targets for common actions/reports
Availability	Defined uptime and maintenance strategy
Scalability	Increasing students, terms, documents and concurrent users
Accessibility	Keyboard access, labels and readable workflows
Compatibility	Supported browsers/devices/network constraints
Maintainability	Versioned code, migrations, tests and documentation
Recoverability	Documented backup/restore/DR
Observability	Metrics, logs, alerts
Data integrity	Transactions, constraints, reconciliation, controlled corrections

36. Testing & Security Assurance

Test area	Examples
Admissions	Incomplete/duplicate applications, rejected/replaced documents, deadlines
Enrollment	Missing prerequisite, full section, conflict, concurrent registration
Grades	Missing/incomplete/correction/repeat/unauthorized change
Curriculum	Version change, substitution, shifting, exceptions
Graduation	Missing unit/prerequisite/internship/clearance
TOR	History, correction, generation, secure release
Authorization	Cross-student access, escalation, API manipulation, RLS failures
Security	Injection, XSS/CSRF, rate limits, session/secret exposure
Recovery	Database restore, document recovery, transaction recovery
Integrations	Duplicate webhook, timeout, provider failure, reconciliation

Use dependency/static analysis, dynamic testing, authorization tests, database-policy tests and periodic penetration testing based on institutional risk.

37. Migration & Deployment

Migration: source inventory → field mapping → cleansing → duplicate review → staging import → validation → reconciliation → approval → production migration.

Keep migration scripts and traceability to source records.

Deployment: infrastructure → database migrations → backend → frontend → integrations → smoke tests → monitoring → controlled release; maintain rollback plans.

38. Reports, Dashboards & Search

Reports: applications, missing requirements, enrollment, capacity, prerequisite deficiencies, missing grades, standing, graduation deficiencies, internship, clearance, TOR, credentials and audit activity.

Provide permission-aware global search. Log and control bulk exports.

Support scheduled reports with explicit recipients.

39. Roles & Permissions

Role	Typical scope
Student/Applicant	Own application, permitted records, planning, enrollment and requests
Instructor	Assigned sections, attendance, assessments, working grades, submission
Adviser	Advising and degree audit within scope
Program Head/Department	Program, section, curriculum and academic approvals
Admissions	Applications and admission workflows
Registrar	Official records, enrollment/grade posting, credentials
Records Officer	Controlled records processing
Finance	Charges, payments, reconciliation, holds
Scholarship Officer	Awards and conditions
OJT/Research Coordinators	Placement/completion or research workflows
Academic Admin	Institutional academic configuration
System Admin	Technical configuration; no automatic unrestricted academic editing
Auditor/Compliance	Read-only/limited audit access
Super Admin	Exceptional platform administration with strong controls

40. Core Business Rules

- Official enrollment requires configured eligibility conditions.
- Prerequisites/corequisites are checked before registration unless an authorized exception exists.
- Section capacity cannot be exceeded without authorized override.
- Schedule conflicts are detected before registration.
- Official grades cannot be silently overwritten.
- Grade corrections preserve original value and complete approval/audit history.
- Curriculum audits use applicable curriculum version and ruleset.
- Graduation approval requires configured completion conditions.
- TOR generation uses authoritative academic records.
- Credential releases are logged.
- Personal/academic records are limited by role and legitimate scope.
- Sensitive operations are authorized server-side and, where applicable, at database level.
- External duplicate events must not create duplicate institutional transactions.
- Record corrections require controlled workflows.

41. Document Templates

Configurable templates: admission application/decision, requirements checklist, enrollment/registration, COR where used, billing statement, advising, add/drop/withdrawal, grade submission/correction, transfer-credit evaluation, internship/OJT completion, thesis/capstone approval, graduation application, degree audit, clearance, TOR, certificate of graduation/degree, registrar certifications and credential verification.

42. Development Roadmap

Stage	Deliverables
1. Discovery	Policies, forms, stakeholders, workflows, curricula, grading, registrar procedures
2. Architecture	Database, backend, auth, authorization, audit, storage, integrations
3. Master Data	Students, programs, courses, curriculum, terms, rooms, instructors
4. Admissions	Applications, requirements, evaluation, decisions

Stage	Deliverables
5. Enrollment	Advising, registration, sections, schedules, changes
6. Academic	Attendance, assessments, grades, standing
7. Exceptions	Transfers, shifting, leave, readmission, repeats, substitutions
8. Services	Finance, scholarships, internship, research, notifications
9. Graduation	Audit, clearance, approval, conferment
10. Registrar/TOR	Official records, credentials, verification
11. Security/QA	RLS/RBAC, audit, backups, tests, privacy
12. Migration	Legacy mapping, cleansing, reconciliation, import
13. Deployment	Staging, pilot, production, monitoring, rollback
14. Operations	Support, incidents, recovery tests, improvement

43. MVP vs Full Production Scope

MVP: student master, admissions, programs/curriculum, terms/courses, sections, enrollment, grades, standing, registrar, degree audit, graduation and basic TOR.

Production expansion: database authorization/RLS, MFA, private storage, workflow engine, notifications, payment/integration handling, advanced scheduling, full audit, migration, DR, monitoring, privacy governance and operations.

Do not defer security, authorization, audit and backup foundations merely because they are not visible user features.

44. Production Readiness Checklist

Academic Core

- Student master
- Admissions/documents
- Curriculum/prerequisites
- Advising/degree audit
- Academic calendar
- Enrollment
- Sections/scheduling
- Changes
- Transfer credits
- Attendance/grades
- Standing
- Internship/research
- Graduation/conferment
- TOR/credentials

Security

- Authentication
- MFA for privileged users
- RBAC
- Database authorization/RLS
- Private storage
- Server-only secrets
- Rate limiting

- Audit logs
- Security monitoring
- Security testing

Data & Operations

- Database constraints
- Transaction safety
- Reconciliation
- Backup
- Restore testing
- Disaster recovery
- Migration
- Environment separation
- Monitoring
- Incident response
- Retention

Production

- API versioning
- Notifications
- Payment/integrations
- Deployment/rollback
- Training
- Support
- Documentation
- Performance testing
- Accessibility review
- Production approval

45. Implementation Notes & Next Steps

This guide is a development blueprint, not a substitute for approved institutional policy. Obtain authoritative curriculum versions, grading rules, enrollment policies, graduation requirements, registrar procedures, document formats, retention schedules, privacy/security requirements and applicable regulations before implementation.

First technical deliverables: requirements/policy catalogue; ERD/data dictionary; role-permission matrix; workflow/state diagrams; API specification; security threat model; UI wireframes; deployment architecture; migration plan; test strategy.

Recommended backend baseline: PostgreSQL + secure server-side API + dedicated authentication + database-level authorization/RLS + private storage + workflow engine + audit logging + automated backups and tested recovery.

Definition of done: A feature is complete only when its data model, authorization, validation, workflow, audit history, error handling, tests, monitoring and recovery implications are addressed.

Appendix A — Reference State Models

Applicant: Draft → Submitted → Verification → Evaluation → Accepted/Waitlisted/Denied → Confirmed/Expired

Enrollment: Planned → Validating → Pending Approval → Assessed → Paid/Cleared → Registered → Changed/Dropped/Withdrawn → Closed

Grade: Working → Submitted → Reviewed → Posted → Corrected (with history)

Graduation: Applied → Under Audit → Deficient/Cleared → Approved → Conferred

Credential: Requested → Identity/Eligibility Check → Processing → Generated → Released → Verified/Superseded

These states should be configurable to institutional policy and are not universal legal definitions.

Appendix B — Security Boundaries

Browser/mobile client = untrusted input boundary. API/server = authentication, authorization and business-rule boundary.

Database = integrity and row-level authorization boundary. File storage = private document boundary. External integrations = separate trust boundaries. Operations = deployment, secrets, backups and monitoring boundary.

Every boundary should have an explicit trust assumption, control and test.

Appendix C — Final Principle

The Transcript of Records should be the **result** of a trustworthy chain of institutional records—not a document manually assembled at the end: admission evidence → identity → curriculum → enrollment → course participation → assessment → official grades → academic history → degree audit → graduation → degree conferment → credential/TOR.